MOBILE DEVELOPMENT

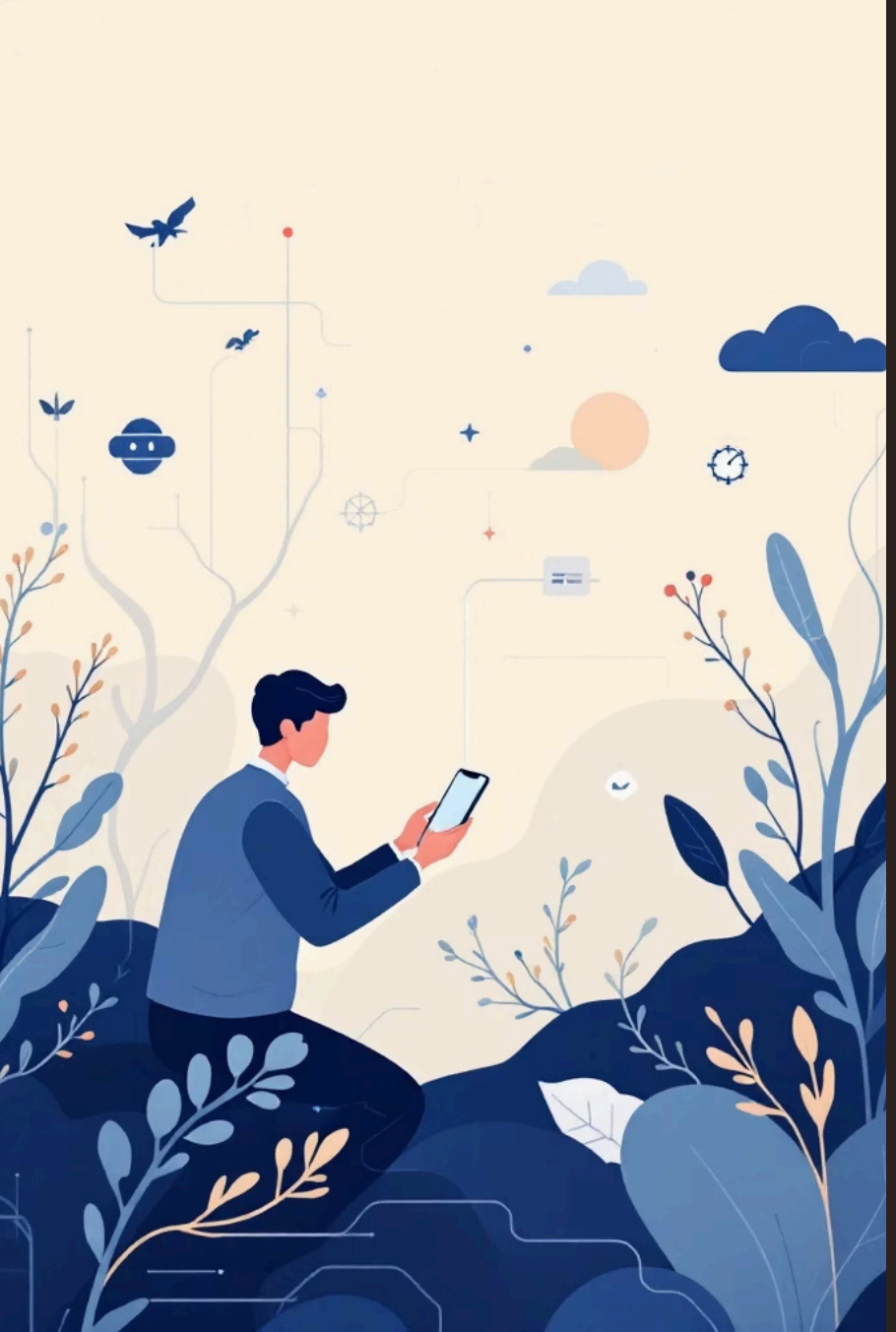
Android / iOS with Flutter

Renel Sainfort
ESILV – Engineering School

CM2: AI in development

SESSION

01 — AI Ecosystem
02 — AI Models & Apps
03 — AI Agentic
04 — AI Prompting
05 — AI Risk & Limit



Ch1apter 01

AI ECOSYSTEM

Context & Industry Impact

FRANCE 2030 — "OSEZ L'IA" PLAN

France's plan to bring AI to every business by 2030 – training, funding, adoption at scale

€109B in private investment attracted to France

3 Ambitions:

- Train 40,000 to 100,000 students per year in AI
- AI adoption goal: from 8% to 50%+ of businesses using AI by 2030
- Make France a European leader in AI

Sources: Government France 2030 & Osez l'IA Plan (2025)

AI is Transforming Work

Healthcare
Diagnosis, drug discovery, prevention

Finance
Fraud detection, trading, personalized advisory

Education
Adaptive learning, AI training

IT/Industry
Cybersecurity, maintenance

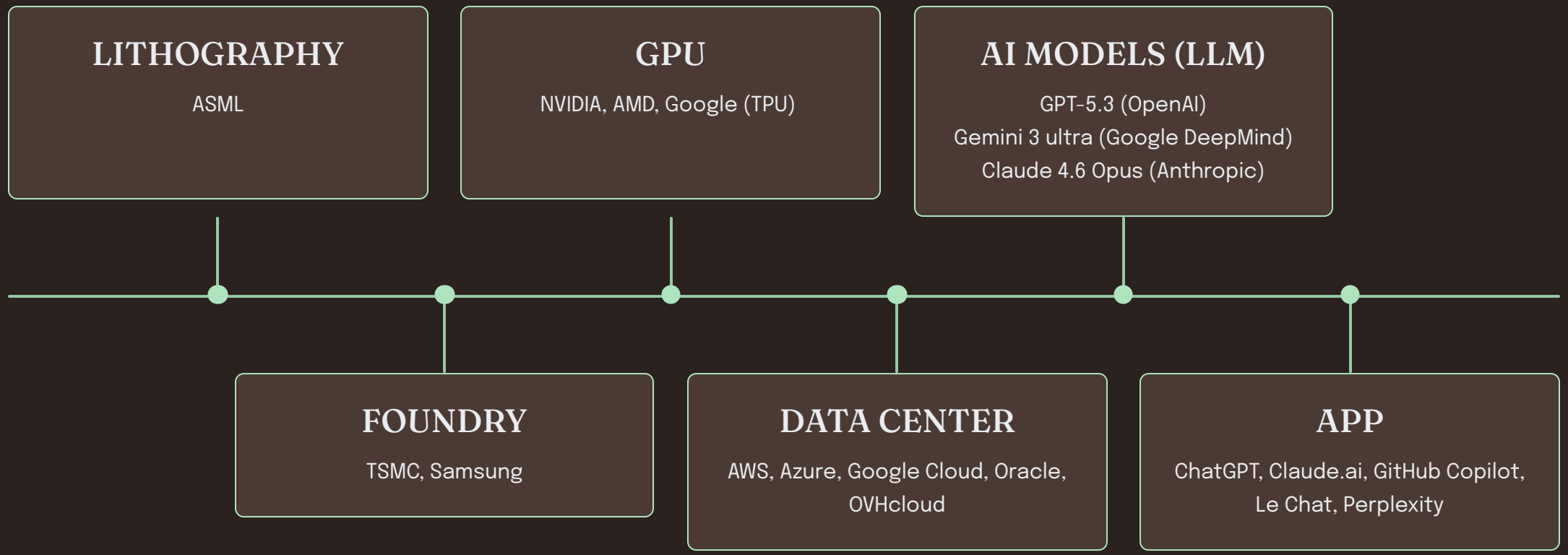
Transition écologique
Sustainable development, green tech, energy transition

Mobilité
Smart transport, electric vehicles, logistics optimization

Robotique
Automation, manufacturing

Agriculture
Precision farming, crop optimization

The AI Value Chain — From Chips to Apps



AI MODELS & APPS

Understanding AI Communication

ANATOMY OF AN AI MODEL

6 keys characteristics that define every LLM

01

Parameters

Number of connections in the model. More capacity, but bigger ≠ smarter.

02

Context Window

Amount of text the model can process in a single conversation.

03

Tokens

The basic unit of measurement for AI. Input and output are counted in tokens.

04

Multimodality

Ability to understand and generate text, images, audio, video and code.

05

Training vs Inference

Training = the model learns. Inference = the model answers.

06

Open vs Closed

Closed = API access only. Open = downloadable weights and code.

Comparative Analysis of Leading AI Models

A breakdown of key characteristics for cutting-edge LLMs

Characteristic	OpenAI : GPT-5.3-Codex 🇺🇸	Google : Gemini 3 Deep Think 🇺🇸	Anthropic : Claude Opus 4.6 🇺🇸	xAI : Grok 4.1 🇺🇸
01. Parameters	~2-5T	~3T	~1.5T	~2T
02. Context Window	512K	1M input	1M	2M
03. Tokens	~0.75 word EN; $2.5/M input	~0.75 word; Google Cloud $	~0.75 word; $15/M output	~0.75 word; xAI API
04. Multimodality	Text+code+image+ workflows	Text+image+video+ math	Text+image+docs+ team agents	Text+image+ real-time
05. Training vs Inference	Training $B; Inference +fast	Training TPU; Iterative Inference	Training AWS; Inference coding	Training xAI; Ultra-fast Inference
06. Open vs Closed	Closed	Closed	Closed	Closed (xAI API)

Comparative Analysis of Leading AI Models

A breakdown of key characteristics for cutting-edge LLMs (Cont.)

Characteristic	Meta : Llama 4 Scout 🇺🇸	Mistral : Magistral Large 2 🇫🇷	DeepSeek : V3.2-Speciale 🇨🇳	Alibaba : Qwen 3.5 🇨🇳
01. Parameters	109B total	123B	Undisclosed	235B (multimodal)
02. Context Window	10M tokens	128K tokens	512K	128K
03. Tokens	~0.75 word; Free open-source	~0.75 word; €2-8/M (API)	~0.75 word;	~0.75 word; Alibaba Cloud
04. Multimodality	Text+image+video (native)	Text+code	Text+agent reasoning	Text+image+audio +video
05. Training vs Inference	Meta Training; single GPU Inference	Mistral Training; Optimized Inference	Scalable RL Training; Efficient Inference	Alibaba Training; Hybrid Thinking
06. Open vs Closed	OPEN	OPEN	OPEN	OPEN

THE APPS

How we actually use these models every day

01

ChatGPT

Powered by GPT-5.2. General-purpose chat, most used worldwide.

02

Claude.ai / Claude Code

Powered by Claude 5.1. Reasoning, coding, long-form analysis.

03

GitHub Copilot

Powered by GPT + Claude. Code assistance integrated in VS Code.

04

Cursor

Multi-model. Complete IDE with integrated AI.

05

Perplexity

Mix LLM + web search. AI search engine.

06

Gemini App

Powered by Gemini 3. Integrated with Google Workspace.

AI AGENTIC

Autonomous Systems

AGENTIC CODING

An AI agent that can autonomously manage and execute development tasks from start to finish.

01

Define the Goal

You provide the high-level objective for the agent.

02

Read Project & Plan

The agent understands the existing codebase and strategizes tasks.

03

Write Code

Generates code based on the plan, adapting to project specifics.

04

Execute & Test

Runs the code, performing checks and validating functionality.

05

Read Errors & Debug

Analyzes output for errors, identifying root causes and solutions.

06

Iterate Autonomously

Applies fixes and repeats the cycle until the goal is achieved.

3 PRINCIPLES OF AGENTIC CODING

What an AI agent needs to work autonomously

ENVIRONMENT

📁 Desktop

📁 Project folder

📁 GitHub repo

PERMISSIONS

👁️ Read

✏️ Write

⚡ Execute

🗑️ Delete

TOOLS

💻 **Local**

Terminal, Git, Compiler

🌐 **Connected**

Web search, APIs

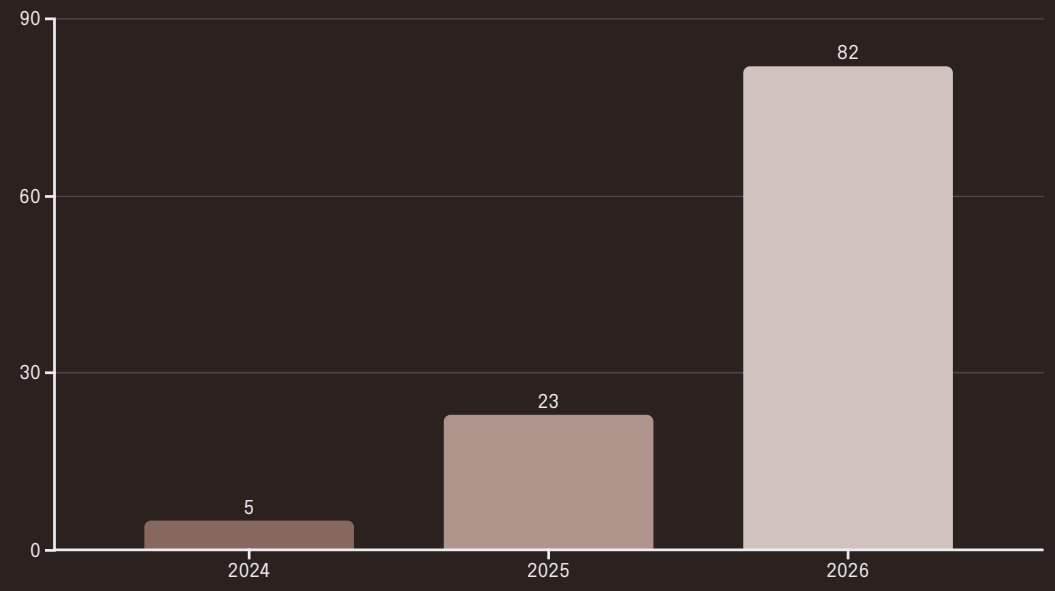
🔌 **Services**

Stripe, Notion, Gmail, Firebase

ADOPTION TRENDS

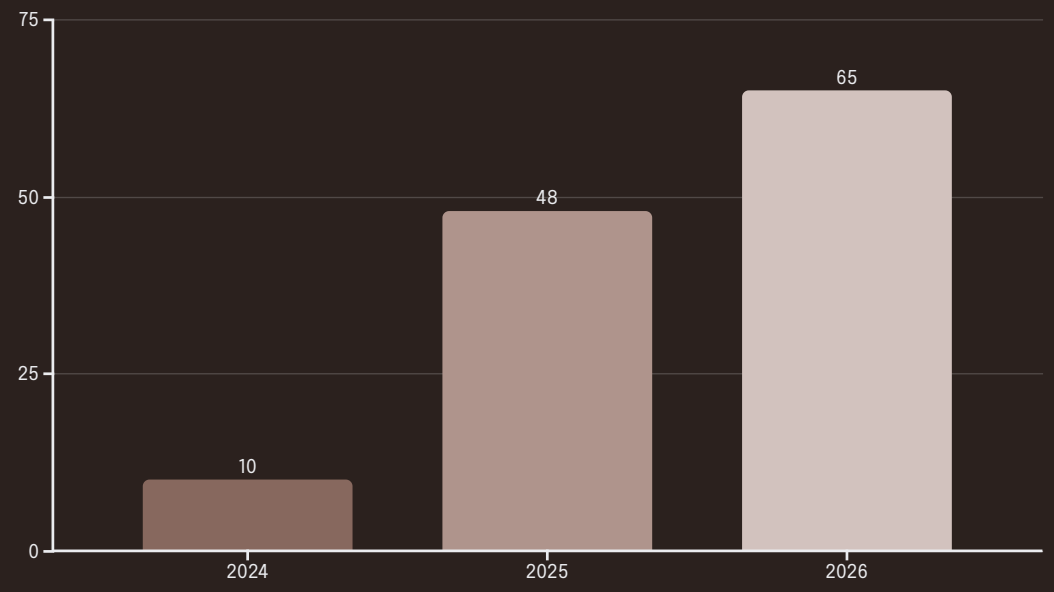
The rapid rise of AI-powered development

Agentic Coding Adoption — World (Developers)



Sources: GitHub, Jellyfish, Anthropic

Agentic Coding Adoption — France (Enterprises).



Sources: Insee, ABBYY, CRÉDOC

SINGLE-MODEL AGENTS

One LLM, maximum performance

01

Claude Code

Strengths

- Peak performance
- Minimal cost (direct API, no intermediary fees)
- Deep integration
- Ultra-simple setup

02

Gemini Code Assist

Limitations

- Locked to one family
- Vendor lock-in

MULTI-MODEL PLATFORMS

Switch models depending on the task

01

Cursor

GPT / Claude / Llama. Full IDE with AI built-in.

Strengths

- Total flexibility – Best model per task
- Switch in 1 click
- Independence
- Testing new LLMs

02

GitHub Copilot

GPT / Claude / Gemini. Embedded in VS Code.

Limitations

- Lite performance –
- Complex setup
- High platform commission fees

AI PROMPTING

Mastering Communication with AI

AI PROMPTING SKILL

The ability to formulate structured instructions for an AI model to achieve precise, reliable, and actionable results.

Why it's Strategic

Precision Determines Quality

The quality of AI output is directly linked to the clarity and accuracy of the prompt.

The Differentiator

With identical AI models, superior prompting transforms generic responses into professional deliverables.

THE RCTFC FRAMEWORK

5 Components of a Good Prompt

1

R — Role

Define the AI's expertise and perspective, guiding its output with a specific persona or function.

2

C — Context

Describe the project, the technical stack, and the target audience to provide essential background information.

3

T — Task

State exactly what you need the AI to produce, outlining the main objective clearly and concisely.

4

F — Format

Specify how the output should be structured and delivered, including desired file types, length, or stylistic elements.

5

C — Constraints

Set the boundaries: define specific rules, limitations, standards, or non-negotiable requirements for the AI.

❌ THE BASIC PROMPT

YouTube

Test bad prompt

Prompt:

"Make me a Flutter app that displays a list of recipes. The app should run from main.dart."

✅ THE RCTFC PROMPT

1

R — Role

"You are an experienced mobile app developer. You write clean, well-organized code with a modern and professional visual style."

2

C — Context

"I'm building a recipe app called **ChefMate**. The users are home cooks aged 25-40 who want to find recipes quickly. The app uses **Flutter**."

3

T — Task

"Create a **recipe list screen** that shows recipe cards with: a photo, the recipe name, the cooking time, the difficulty level, and a favorite button. Add a **search bar** at the top and **category filters**: All, Breakfast, Lunch, Dinner, Dessert."

4

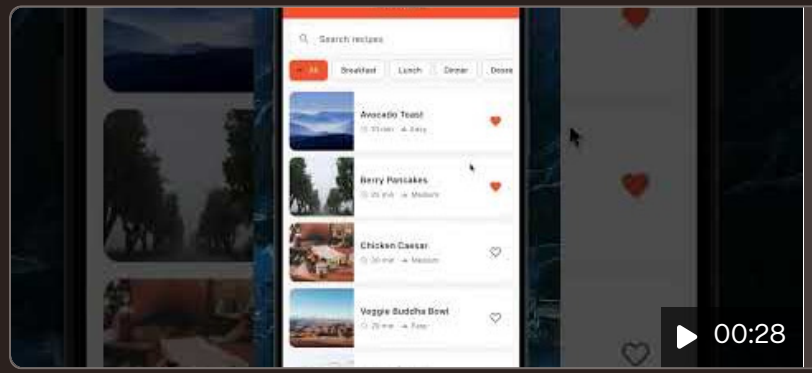
F — Format

"Give me a **complete app** that I can run directly. One single file with comments explaining each section. Use **sample data** with at least 6 recipes."

5

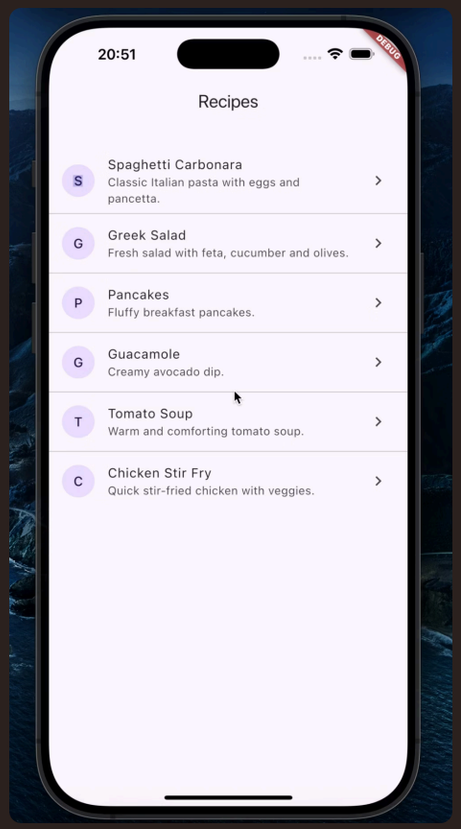
C — Constraints

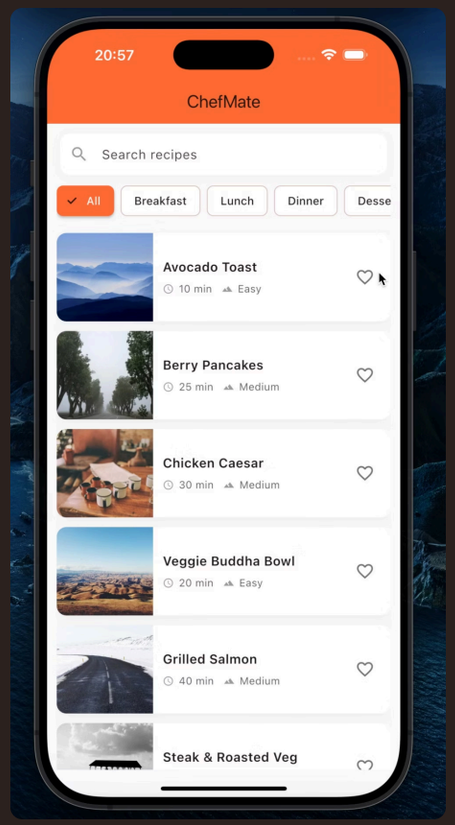
"Main color: coral #FF6B35. Background: light grey #FAFAFA. Cards: white with rounded corners and a subtle shadow. The list must scroll. No external libraries."



YouTube

test good prompt


20:51
DEBUG
Recipes
S
Spaghetti Carbonara
Classic Italian pasta with eggs and pancetta.
G
Greek Salad
Fresh salad with feta, cucumber and olives.
P
Pancakes
Fluffy breakfast pancakes.
G
Guacamole
Creamy avocado dip.
T
Tomato Soup
Warm and comforting tomato soup.
C
Chicken Stir Fry
Quick stir-fried chicken with veggies.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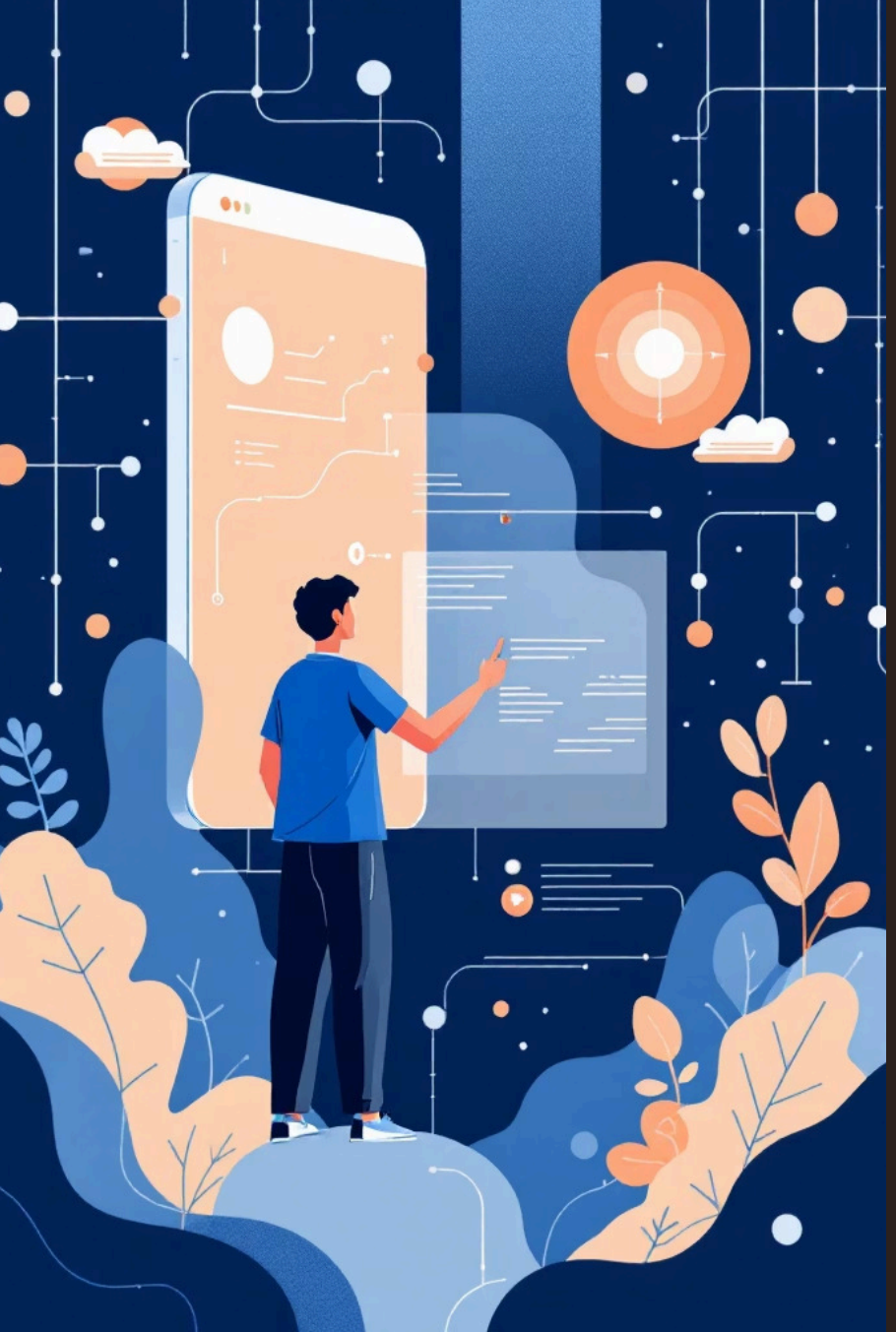
ChefMate
Search recipes
All
Breakfast
Lunch
Dinner
Desse
Avocado Toast
10 min
Easy
Berry Pancakes
25 min
Medium
Chicken Caesar
30 min
Medium
Veggie Buddha Bowl
20 min
Easy
Grilled Salmon
40 min
Medium
Steak & Roasted Veg



AI RISK & LIMIT

Understanding Challenges & Responsibilities

AI RISKS & LIMITS

Critical Challenges in AI Adoption

🔒 Security Concerns

- **Hallucinations:**
- **Vulnerable Code:**
- **Data Leaks:**

⚡ Energy Footprint

META's Altoona data center: **1,401 MW** – more than all of Paris intra-muros (2.2M people).

🏛️ Societal Impact

- **Tech Concentration:** A few companies control models, data & infrastructure
- **Two-Speed World:** Divide between AI creators and AI dependents

Sources: BlackRidge Research, INSEE

WHAT'S NEXT

TP2

Training RCTFC

CM2

Flutter + Doc

QUESTIONS ?